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Dividend Momentum and Stock Return Predictability: A Bayesian Approach

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1 Big picture

- **Question.** Can aggregate stock return predictability be estimated in a way that is economically disciplined, respects the Campbell–Shiller identity, and still works out of sample?
- **Core problem.** Classical VARs with flat priors can imply excessive return predictability because persistent valuation ratios and small-sample bias make predictive coefficients look too strong.
- **Core innovation.** The paper develops Bayesian methods that impose Campbell–Shiller cross-equation restrictions while also using informative priors that express skepticism about return predictability.
- **Key economic mechanism.** Dividend growth is persistent after controlling for the price-dividend ratio. This creates **dividend momentum**: positive dividend-growth shocks raise current dividends, current returns, and expected future returns.
- **Why dividend momentum matters.** It is an additional predictability channel that is missed when dividend growth is dropped from the VAR or treated as redundant.
- **Empirical setting.** The main VAR uses annual U.S. postwar S&P 500 data from 1947 to 2024 for log dividend growth, the log price-dividend ratio, and log market returns.
- **Main in-sample result.** The restricted informative posterior is skeptical about one-period return predictability, but it still finds economically meaningful long-horizon predictability because price-dividend mean reversion and dividend momentum shape future expected returns.
- **Main out-of-sample result.** The baseline restricted informative prior is the only approach in Table 3 that beats the naive historical-average forecast across all reported horizons; out-of-sample R^2 peaks around five years at 28.8%.
- **Portfolio result.** Long-horizon asset allocation based on the restricted informative posterior delivers better risk-adjusted wealth than flat-prior VARs and avoids implausibly large leverage or short positions.
- **Quarterly robustness.** A state-space model with seasonal dividend components shows that dividend momentum survives at quarterly frequency and is linked to macroeconomic boom–bust dynamics.
- **Takeaway.** Return predictability is neither zero nor as strong as flat-prior VARs suggest. The useful middle ground comes from combining Bayesian shrinkage with the accounting discipline of the Campbell–Shiller identity.

2 Data and measurement

2.1 Core annual variables

The baseline state vector is

$$y'_{t+1} = [\Delta d_{t+1}, pd_{t+1}, r_{t+1}],$$

where

- Δd_{t+1} is log dividend growth;
- pd_{t+1} is the log price-dividend ratio;
- r_{t+1} is the log market return.

The main annual sample is U.S. postwar S&P 500 data from 1947 to 2024. Dividends are measured without reinvestment because reinvested-dividend measures can distort dividend growth dynamics.

Interpretation: The paper treats dividends as a state variable rather than as a redundant object. This is essential because persistent dividend growth is exactly what generates dividend momentum.

2.2 Flat-prior benchmark

The paper first estimates the VAR under flat priors without imposing the Campbell–Shiller restrictions. This benchmark shows:

- persistent dividend growth, with $\phi_{d,d}$ around 0.41;
- a highly persistent price-dividend ratio, with $\phi_{pd,pd}$ around 0.91;
- return predictability from the price-dividend ratio, with $\phi_{r,pd} < 0$;
- relatively imprecise estimates of the steady-state means.

Interpretation: The flat-prior estimates reveal the empirical pattern, but they are not the preferred inference because they can exaggerate predictability and understate persistence in valuation ratios.

2.3 Out-of-sample forecasting design

The out-of-sample exercise begins in 1973:Q1 and ends in 2024:Q4. Each investor or forecasting model is re-estimated recursively using an expanding window. Forecasts are evaluated for cumulative excess returns at horizons $h = 1, \dots, 10$.

Interpretation: The design asks whether the in-sample predictability is useful for real-time investors, not merely whether coefficients are significant in the full sample.

2.4 Quarterly extension

The quarterly extension explicitly models seasonality in dividends. The authors decompose dividend growth and the price-dividend ratio into seasonal and seasonally adjusted components, estimate the model in state-space form, and recover latent seasonally adjusted dividend growth.

Interpretation: This addresses the concern that annual dividend-growth persistence is just hidden seasonality. The quarterly results show that dividend momentum remains after modeling seasonal components directly.

3 Models and empirical specifications

3.1 Baseline VAR

The paper begins with a first-order VAR:

$$\begin{bmatrix} \Delta d_{t+1} \\ pd_{t+1} \\ r_{t+1} \end{bmatrix} = \begin{bmatrix} c^d \\ c^{pd} \\ c^r \end{bmatrix} + \begin{bmatrix} \phi_{d,d} & \phi_{d,pd} & \phi_{d,r} \\ \phi_{pd,d} & \phi_{pd,pd} & \phi_{pd,r} \\ \phi_{r,d} & \phi_{r,pd} & \phi_{r,r} \end{bmatrix} \begin{bmatrix} \Delta d_t \\ pd_t \\ r_t \end{bmatrix} + \begin{bmatrix} u_{t+1}^d \\ u_{t+1}^{pd} \\ u_{t+1}^r \end{bmatrix}, \quad (1)$$

with innovations $u_{t+1} \sim N(0, \Sigma)$.

Interpretation: This is the smallest VAR that lets dividends, valuation ratios, and returns jointly determine each other.

3.2 Campbell–Shiller identity

The log-linear Campbell–Shiller identity is

$$r_{t+1} \approx \kappa + \rho p d_{t+1} - p d_t + \Delta d_{t+1}. \quad (2)$$

For innovations, this implies

$$u_{t+1}^r = u_{t+1}^d + \rho u_{t+1}^{pd} + \eta_{t+1}. \quad (3)$$

Ignoring the small approximation error gives restrictions on the VAR coefficients:

$$c^r = c^d + \rho c^{pd} + \kappa, \quad (4)$$

$$\phi_{r,d} = \phi_{d,d} + \rho \phi_{pd,d}, \quad (5)$$

$$\phi_{r,pd} = \phi_{d,pd} + \rho \phi_{pd,pd} - 1, \quad (6)$$

$$\phi_{r,r} = \phi_{d,r} + \rho \phi_{pd,r}. \quad (7)$$

It also implies restrictions on covariances:

$$\text{Cov}(u_{t+1}^r, u_{t+1}^d) = \rho \text{Cov}(u_{t+1}^{pd}, u_{t+1}^d) + \text{Var}(u_{t+1}^d), \quad (8)$$

$$\text{Cov}(u_{t+1}^r, u_{t+1}^{pd}) = \rho \text{Var}(u_{t+1}^{pd}) + \text{Cov}(u_{t+1}^d, u_{t+1}^{pd}). \quad (9)$$

Interpretation: Because these restrictions come from an identity, not a behavioral model, they are a powerful source of discipline. The challenge is imposing them in Bayesian VAR estimation without dropping variables.

3.3 Stationarity restriction

The CS derivation requires a stationary price-dividend ratio. The authors impose the multivariate stationarity condition

$$\max\{\text{eig}(\Phi_1)\} < 1. \quad (10)$$

Interpretation: This prevents the Bayesian procedure from assigning probability to parameter values that violate the economic environment behind the CS identity.

3.4 Informative priors

The paper combines a Minnesota prior with a single-unit-root prior. In simplified notation,

$$\text{vec}(\Phi_1) \mid \Sigma \sim N(\text{vec}(\Phi_0), \Sigma \otimes \Omega), \quad (11)$$

where Ω is scaled by a tightness parameter λ . The single-unit-root prior introduces dummy observations that place prior mass on plausible unconditional means and reduce overfitting by deterministic components.

Interpretation: The prior encodes skepticism about short-run return predictability and pushes the price-dividend ratio toward high persistence, avoiding the excessive mean reversion often produced by flat priors.

3.5 Restricted posterior

The methodological contribution is an importance-sampling approach that draws from the posterior while enforcing:

- the CS restrictions on Φ ;
- the CS restrictions on Σ ;
- stationarity of the system.

Interpretation: A standard informative prior does not automatically satisfy the identity. The paper’s restricted posterior combines prior discipline and identity discipline simultaneously.

3.6 Return predictability and R^2

The return equation’s predictability is summarized by R^2 at one period and at long horizons. For out-of-sample forecasts, the paper reports

$$R_{OOS}^2(h) = 1 - \frac{\sum_t (R_{t,t+h} - \hat{R}_{t,t+h})^2}{\sum_t (R_{t,t+h} - \bar{R}_{t,t+h})^2}. \quad (12)$$

Interpretation: Positive R_{OOS}^2 means the model beats the naive historical-average benchmark in real-time forecasting.

3.7 Cash-flow and discount-rate news

The Campbell–Shiller decomposition writes unexpected returns as

$$r_{t+1} - E_t r_{t+1} = NCF_{t+1} - NDR_{t+1}, \quad (13)$$

where NCF_{t+1} is cash-flow news and NDR_{t+1} is discount-rate news.

Interpretation: Dividend momentum means that dividend-growth shocks affect not only cash-flow news but also expected future returns, linking cash-flow shocks and discount-rate news.

3.8 Long-horizon variance

For cumulative returns $r_{t,t+k} = \sum_{j=1}^k r_{t+j}$, the paper studies

$$\frac{\text{Var}_t(r_{t,t+k})}{k \text{Var}_t(r_{t,t+1})}. \quad (14)$$

Interpretation: Price-dividend mean reversion lowers long-horizon variance, creating a positive hedging demand for long-run investors. Dividend momentum raises long-horizon variance because dividend-growth shocks raise future expected returns, partially offsetting that hedging motive.

4 Identification and empirical design

4.1 What makes the approach credible

- The CS restrictions are accounting restrictions implied by the definition of returns.
- The Bayesian priors are chosen to be skeptical about return predictability rather than to inflate it.
- The out-of-sample design evaluates forecasts recursively rather than using full-sample hindsight.
- The quarterly extension explicitly models seasonality instead of assuming it away.
- The portfolio tests ask whether predictability leads to economically plausible allocation gains, not just statistical fit.

Interpretation: The paper’s strength is that it combines theory, prior skepticism, and out-of-sample performance. Each component alone is insufficient.

4.2 Why dividend growth cannot simply be omitted

A common practice is to drop dividend growth from the VAR and infer it from the CS identity. The paper argues this can hide dividend momentum because it imposes extra restrictions on dividend dynamics and makes dividend growth unable to act as an independent predictor.

Interpretation: Omitting dividend growth is not innocuous when $\phi_{d,d} \neq 0$. The missing state variable changes both return forecasts and the risk perceived by long-run investors.

4.3 Main limitations

- The results depend on prior choices, although the paper chooses hyperparameters by marginal likelihood and checks sensitivity.
- The annual sample is relatively short for estimating long-horizon predictability.
- The asset allocation analysis uses stylized investors and model-implied forecasts.
- The quarterly extension requires a state-space treatment of seasonality and stochastic volatility.

Interpretation: The paper is not claiming that stock returns are easy to forecast. It claims that a disciplined Bayesian system can recover a moderate and economically useful amount of predictability.

5 Tables and figures

The figures and tables below are directly cropped from the original paper and grouped to save space.

5.1 Figure 1 + Table 1: CS restrictions and the flat-prior benchmark

Read:

- Figure 1 shows that likelihood draws align with the CS restrictions, but standard informative priors and the resulting unrestricted posteriors do not.
- The gap is especially visible for the covariance restrictions.
- Table 1 reports the flat-prior posterior. Dividend growth is persistent, the price-dividend ratio is highly persistent, and the price-dividend ratio predicts returns.
- The steady-state means are imprecisely estimated under flat priors.

Economic message: Flat priors reveal the data pattern but do not deliver a disciplined posterior. The paper’s main method is needed because a useful prior must also respect the accounting identity.

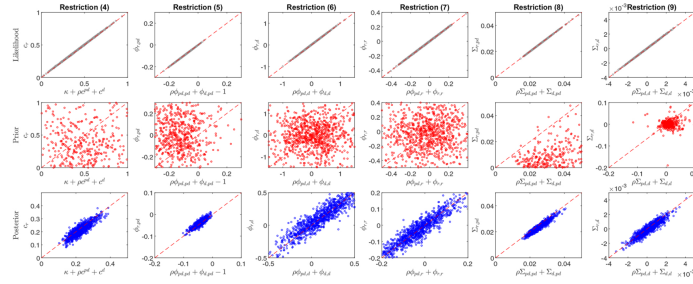


Figure 1
CS Restrictions for Φ and Σ under informative priors
Each row of panels plots random draws from the likelihood, prior, and posterior of the VAR parameters, respectively. Each column corresponds to one CS restriction (4)–(9). Close alignment along the 45-degree line indicates the CS restrictions are tightly satisfied. Likelihood draws (top panels) respect the restrictions. Standard priors (middle panels) do not satisfy the restrictions, and neither does the resulting posterior (lower panels). Red represents the prior, gray is the likelihood, and blue is the posterior.

Figure 1: CS restrictions for Φ and Σ .

Table 1
Posterior distribution of μ , Φ_1 , and Σ under flat priors

μ	Φ_1		
	Δd_t	pd_t	r_t
Δd_{t+1}	0.056 [0.034, 0.073]	0.409 [0.310, 0.510]	0.004 [−0.009, 0.018]
pd_{t+1}	3.846 [3.447, 4.469]	−0.371 [−0.682, −0.058]	0.930 [0.887, 0.973]
r_{t+1}	0.079 [0.042, 0.103]	0.056 [−0.223, 0.342]	−0.092 [−0.131, −0.053]
Σ (corr/std)			
	u_{t+1}^d	u_{t+1}^{pd}	u_{t+1}^r
u_{t+1}^d	0.053 [0.049, 0.058]		
u_{t+1}^{pd}	−0.329 [−0.429, −0.227]	0.168 [0.156, 0.183]	
u_{t+1}^r	−0.003 [−0.114, 0.110]	0.945 [0.932, 0.957]	0.154 [0.143, 0.168]

The table shows the parameter estimates under a flat prior for a first-order VAR model including a constant, the log dividend growth (Δd_{t+1}), the price-dividend ratio (pd_{t+1}), and the log market return (r_{t+1}). For each coefficient, the first line reports the posterior median value, and the second line reports the 68th posterior credible intervals in square brackets. The table also reports the parameters of the correlation matrix of the innovations with innovation standard deviations on the diagonal, labeled “corr/std,” instead of the parameters of the covariance

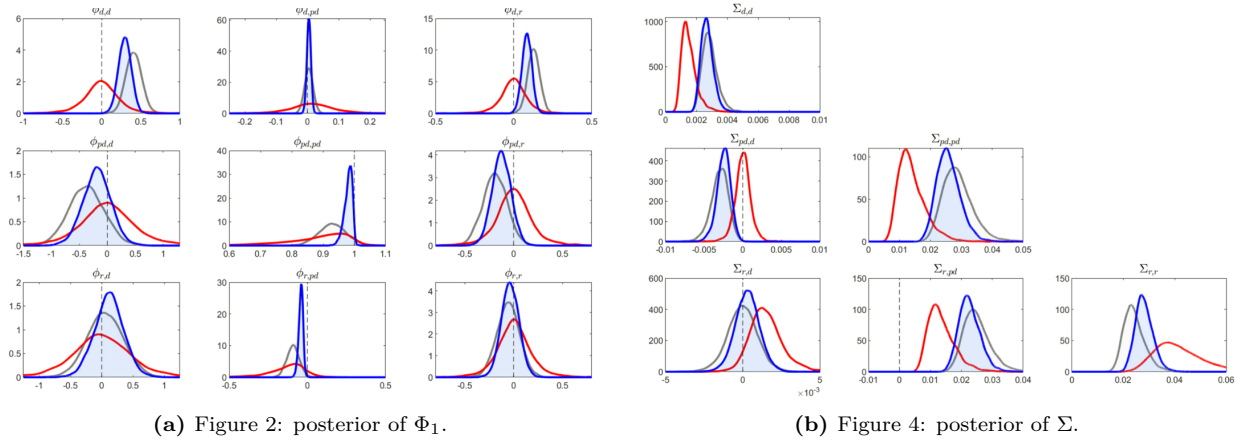
Table 1: Flat-prior posterior for μ , Φ_1 , and Σ .

5.2 Figures 2–4: restricted informative posterior

Read:

- Figure 2 shows shrinkage and CS restrictions in the slope matrix Φ_1 .
- The posterior keeps $\phi_{d,d} > 0$, confirming persistent dividend growth.
- $\phi_{pd,pd}$ becomes very persistent, closer to nonstationarity than the flat-prior likelihood suggests.
- Figure 3 shows that priors matter for the unconditional means, where the likelihood is weak.
- Figure 4 shows how CS restrictions reshape the covariance matrix, especially covariances involving return shocks.

Economic message: The posterior is neither pure data nor pure prior. It combines evidence of dividend momentum with skepticism about short-run return predictability.



(a) Figure 2: posterior of Φ_1 .

(b) Figure 4: posterior of Σ .

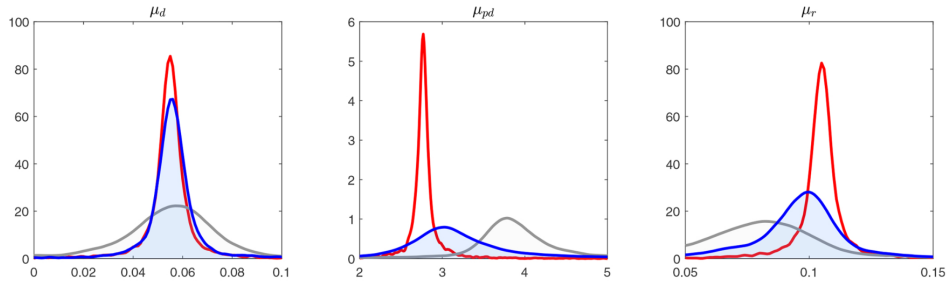


Figure 3
Bayesian restricted prior and posterior of μ .
Each panel displays the prior, likelihood, and posterior probability density functions for each of the parameters in μ . Red represents the prior, gray is the likelihood, and blue is the posterior.

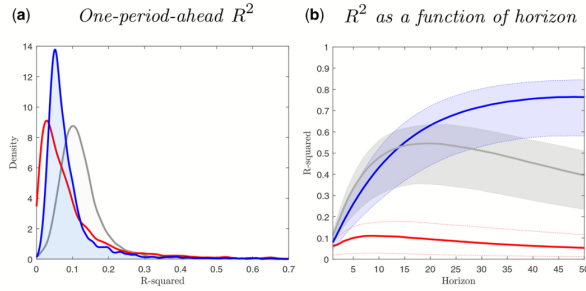
Figure 3: posterior of unconditional means μ .

5.3 Figures 5–6 + Table 2: predictability, expected returns, and news

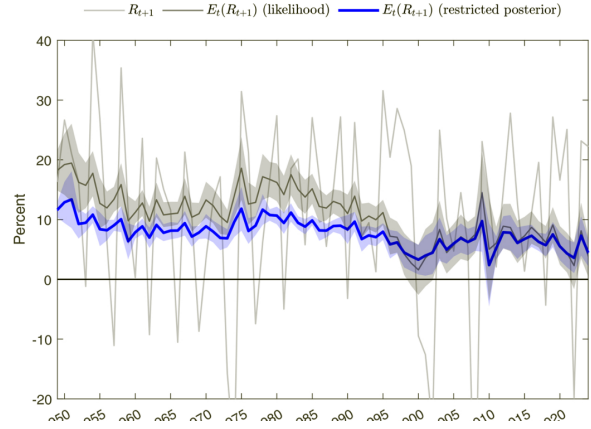
Read:

- Figure 5 shows that the restricted informative posterior assigns low short-run return predictability but meaningful long-horizon predictability.
- Figure 6 shows expected returns are time varying but more stable and never strongly negative under the restricted informative posterior.
- Table 2 decomposes cash-flow and discount-rate news. Price-dividend shocks explain most discount-rate-news variance, while dividend-growth shocks explain most cash-flow-news variance.
- The news components are correlated because dividend momentum links dividend shocks to future expected returns.

Economic message: The model does not reject return predictability; it relocates the action from excessive one-period predictability to more plausible long-horizon predictability.



(a) Figure 5: return equation R^2 .



(b) Figure 6: one-period expected return.

Table 2
Shock's contribution to $NC F_{t+1}$ and NDR_{t+1}

	Total	u_{t+1}^{pd}	u_{t+1}^d
$\text{Var}(NDR_{t+1})$	0.023 [0.013, 0.035]	97.1% [93.9%, 98.7%]	2.9% [1.3%, 6.1%]
$\text{Var}(NC F_{t+1})$	0.007 [0.005, 0.010]	8.5% [0.7%, 32.5%]	91.3% [67.3%, 99.0%]
$\text{Corr}(NDR_{t+1}, NC F_{t+1})$	16.1% [-25.9%, 55.1%]	-1.9% [-43.6%, 38.9%]	15.9% [10.5%, 22.9%]

We report the posterior median and the 68th posterior credible intervals. The "Total" column reflects the posterior of moments, while the u_{t+1}^{pd} and u_{t+1}^d columns reflect the posterior contribution of the two shocks.

Table 2: shock contributions to $NC F_{t+1}$ and NDR_{t+1} .

5.4 Figures 7–8: dividend momentum and long-run risk

Read:

- Figure 7 compares a no-dividend-momentum case with a dividend-momentum case.
- With $\phi_{d,d} = 0$, dividend-growth shocks do not change future expected dividends or returns beyond impact.
- With $\phi_{d,d} > 0$, dividend-growth shocks raise both future dividends and future expected returns.

- Figure 8 shows that dividend momentum raises the variance ratio of long-horizon returns relative to a case with only price-dividend mean reversion.

Economic message: Dividend momentum adds long-horizon risk. It weakens the standard long-run hedging argument that stocks become much safer at long horizons solely because valuation ratios mean-revert.

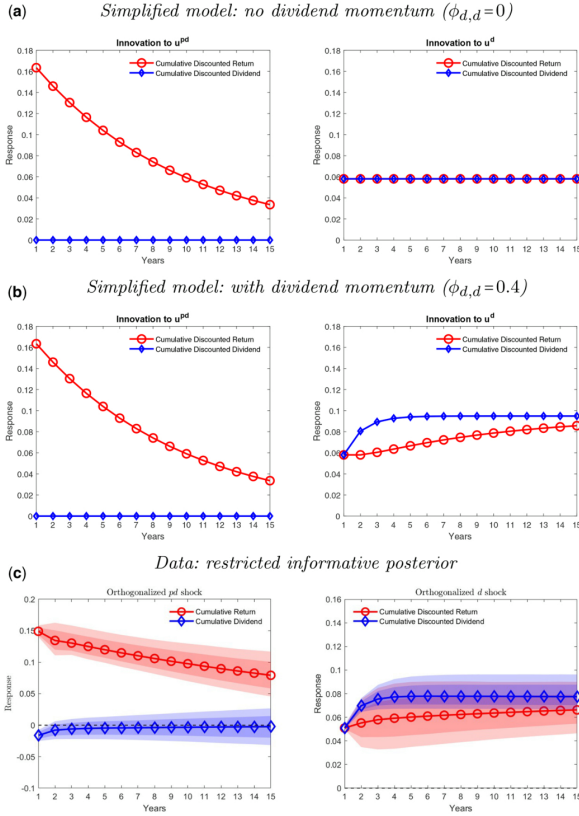


Figure 7

(a) Figure 7: IRFs and dividend momentum.

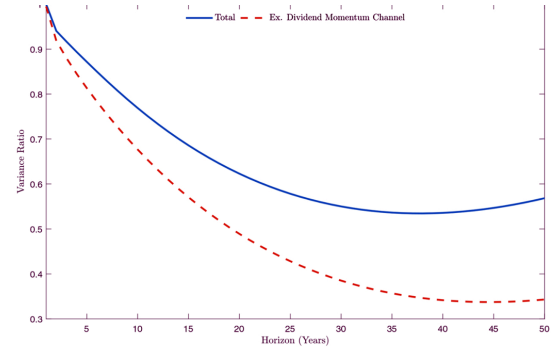


Figure 8

(b) Figure 8: long-horizon variance ratio.

5.5 Table 3 + Figures 9–10: out-of-sample performance and long-term asset allocation

Read:

- Table 3 shows that flat-prior approaches produce negative out-of-sample R^2 across horizons.
- Informative priors without CS restrictions improve performance but generally do not beat the naive benchmark.
- The baseline restricted informative prior beats the naive benchmark at every horizon, peaking at 28.8% at $h = 5$.
- Figure 9 shows that the restricted informative-prior investor accumulates more risk-adjusted wealth with more moderate equity weights.
- Figure 10 decomposes gains: most come from restrictions on Φ , with additional gains from restrictions on Σ .

Economic message: The paper’s empirical punchline is out of sample. Neither shrinkage alone nor CS restrictions alone are enough; the combination is what delivers useful forecasts and allocations.

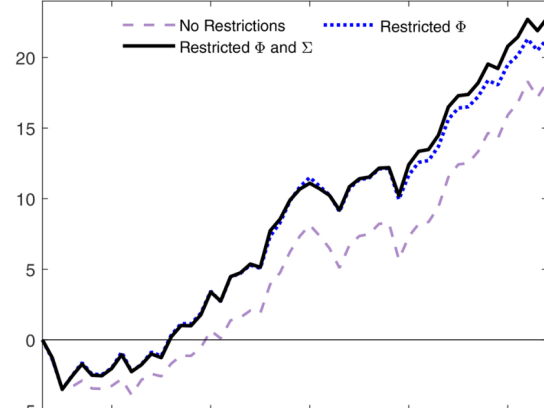
Table 3

Out-of-sample return equation R^2

	Flat priors Omitting divs.	Including divs.	Informative priors Unrestricted (inc. divs.)	Unrestricted (omit. divs.)	Baseline
$h=1$	-9.7%	-12.0%	-1.1%	-0.4%	5.9%
$h=2$	-7.9%	-10.5%	0.2%	0.8%	14.5%
$h=3$	-26.9%	-28.8%	-0.9%	0.2%	21.5%
$h=4$	-39.9%	-39.1%	-4.8%	-3.3%	26.2%
$h=5$	-44.8%	-41.7%	-13.6%	-12.3%	28.8%
$h=6$	-82.5%	-76.0%	-37.2%	-36.2%	27.0%
$h=7$	-126.3%	-115.3%	-60.2%	-59.2%	22.3%
$h=8$	-161.1%	-145.9%	-80.8%	-79.7%	19.7%
$h=9$	-210.4%	-190.4%	-104.6%	-103.5%	15.5%
$h=10$	-292.4%	-264.9%	-138.5%	-137.0%	8.5%

Percentage improvement in out-of-sample fit of each investor with respect to the naive investor. Out-of-sample

(a) Table 3: out-of-sample return R^2 .



(b) Figure 10: wealth-gain decomposition.

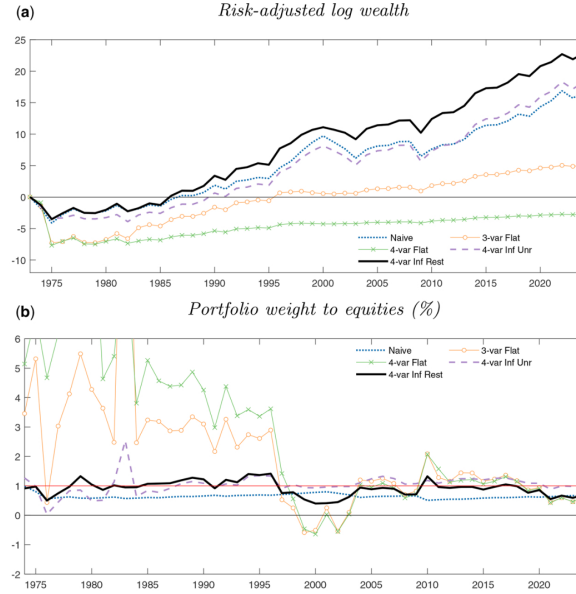


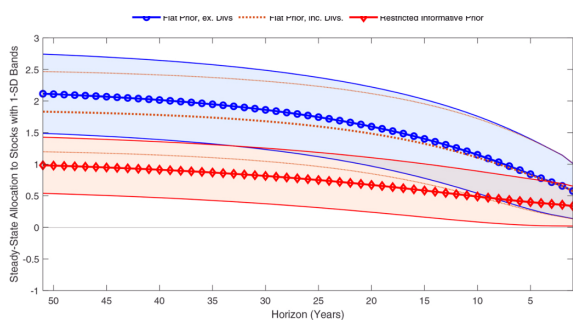
Figure 9: long-term asset allocation strategy.

5.6 Figures 11–14: allocation slopes and prior tightness

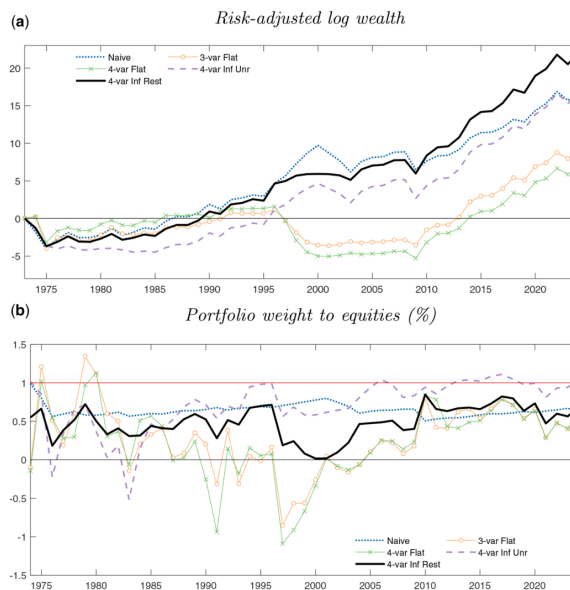
Read:

- Figure 11 shows that the restricted informative prior implies more realistic stock allocations over long horizons.
- Flat-prior investors can demand very high leverage because they infer too much mean reversion and too much predictability.
- Figures 12 and 14 show that the baseline hyperparameter choice is close to Sharpe-ratio maximizing regions, both for long-horizon and myopic strategies.
- Figure 13 shows that the myopic strategy also benefits from the restricted informative prior but with less dramatic allocation dynamics.

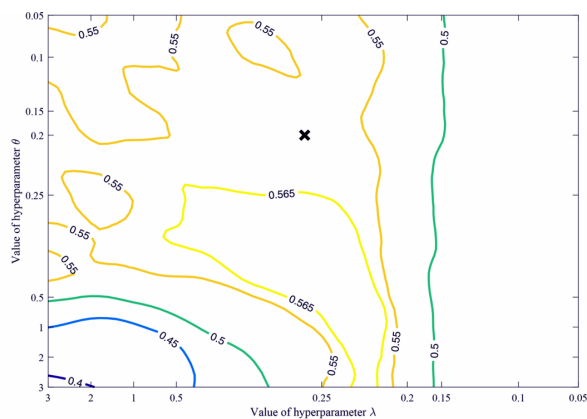
Economic message: Prior tightness is not just a statistical tuning choice. It directly determines whether the model produces plausible investment advice.



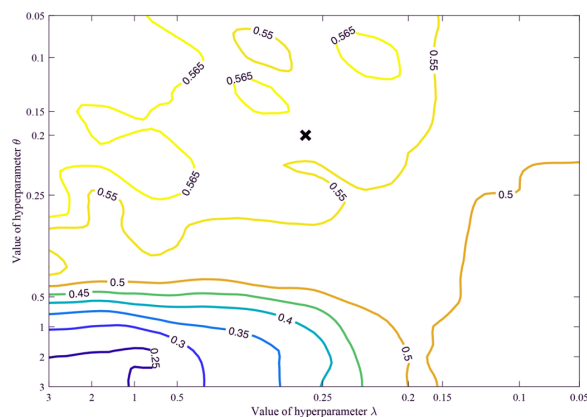
(a) Figure 11: steady-state allocation.



(b) Figure 13: myopic allocation strategy.



(a) Figure 12: long-horizon Sharpe ratio.



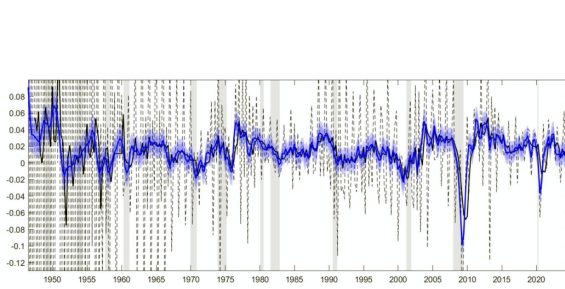
(b) Figure 14: myopic Sharpe ratio.

5.7 Figures 15–17: quarterly data and macroeconomic implications

Read:

- Figure 15 recovers seasonally adjusted quarterly dividend growth and shows that early-sample dividend seasonality was strong but declined over time.
- Figure 16 shows dividend momentum survives in quarterly data: dividend-growth shocks lead to persistent movements in dividends and expected returns.
- Figure 17 shows macroeconomic responses to dividend-growth shocks: consumption, GDP, investment, TFP, and hours display boom–bust dynamics.

Economic message: Dividend momentum is not just an artifact of annual measurement. It is connected to broader macroeconomic dynamics and remains visible after explicitly modeling seasonality.



(a) Figure 15: seasonally adjusted dividend growth.

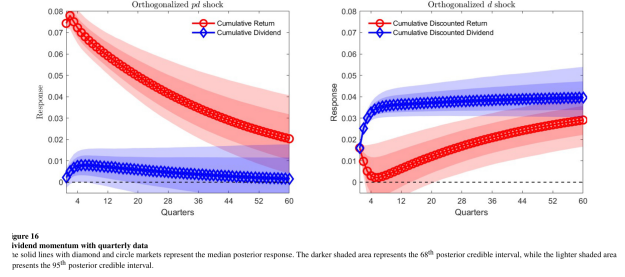


Figure 16: **hidden momentum with quarterly data**
solid lines with diamond and circle markers represent the median posterior response. The darker shaded area represents the 68th posterior credible interval, while the lighter shaded area represents the 95th posterior credible interval.

(b) Figure 16: quarterly dividend momentum.

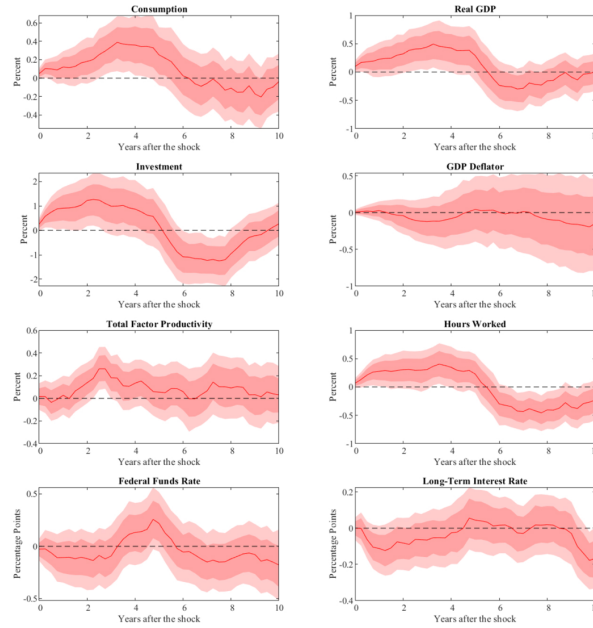


Figure 17
Local projections of macroeconomic variables on dividend growth shock
The solid lines represent the median posterior response. The darker shaded area represents the 68th posterior

Figure 17: local projections of macro variables on a dividend-growth shock.

6 Short summary

- The paper studies aggregate stock return predictability using a Bayesian VAR for returns, dividend growth, and the price-dividend ratio.
- The key methodological contribution is a posterior sampler that imposes Campbell–Shiller restrictions while allowing informative priors.
- Flat priors tend to exaggerate predictability because of persistent valuation ratios and small-sample bias.
- Informative priors encode skepticism about return predictability and shrink implausible parameter values.
- The Campbell–Shiller identity forces returns, dividends, and valuation ratios to be jointly consistent.

- Dividend growth is persistent after controlling for valuation ratios.
- Persistent dividend growth creates dividend momentum: dividend shocks raise future expected dividends and returns.
- Dividend momentum increases long-horizon return variance and partially offsets the hedging demand generated by price-dividend mean reversion.
- The restricted informative posterior yields positive out-of-sample R^2 from 1973:Q1 to 2024:Q4, whereas flat-prior models perform poorly.
- Asset allocation based on the restricted informative posterior generates superior risk-adjusted wealth with more realistic equity weights.
- Quarterly robustness shows dividend momentum survives after modeling seasonality and is linked to macroeconomic boom–bust dynamics.

7 Conclusion

7.1 Core conclusion

Antolín-Díaz, Petrella, and Rubio-Ramírez show that stock return predictability can be economically meaningful without being as extreme as flat-prior predictive regressions imply. The correct discipline is twofold: Bayesian shrinkage prevents excessive predictability, and Campbell–Shiller restrictions ensure that returns, dividends, and valuation ratios remain jointly coherent.

7.2 Economic intuition

The main intuition is dividend momentum. When dividend growth is persistent, a positive dividend-growth shock does more than increase current cash flows. It also changes expectations of future dividend growth and future returns. This creates a procyclical source of expected-return variation that standard models miss when dividend growth is omitted from the state vector.

7.3 Takeaway

The paper’s takeaway is that useful return predictability requires structure. Data alone with flat priors overfits; priors alone can ignore accounting identities; CS restrictions alone can still leave unrealistic forecasts. Combining all three produces moderate predictability, better out-of-sample forecasts, and more plausible portfolio implications.